Starting fresh (21/02/26)

$$V^{\pi}(x) \underset{\text{Distributional level}}{\overset{D}{=}} R + \gamma V^{\pi}(x')$$

Probability density function (PDF)

$$P_V(x) = (P_R * P_{\gamma V'})(x)$$

Convolution (very hard to do convolution on PDFs)

$$C_{V^{\pi}}(w) \triangleq E\left[e^{iwV^{\pi}}\right]$$

Characteristic Function of a random variable

represent the PDF in frequency space using the fourier transform

Goal: Have a Bellman Operator on a Distribution level

↓

In distribution space, this becomes a convolution of their PDF

Fourier Transform $(PDF_X) = CF_X$

↓

In frequency space, this becomes a element-wise multiplication

if r.v $X = R + \gamma V^{\pi}(x')$

$$C_X = E\left[e^{iwX}\right]$$
$$= E\left[e^{iw(R+\gamma V^{\pi}(x'))}\right]$$
$$= E\left[e^{iwR} \cdot e^{iw\gamma V^{\pi}(x')}\right]$$

let's call the CF (pdf in freq space)

$\tilde{V}(w;x) = E\left[e^{iwG^{\pi}(x)}\right]$

$\tilde{V}(\gamma w;x') = E\left[e^{iw\gamma G^{\pi}(x')}\right]$

$$\tilde{V}(w;x) = E\left[e^{iwR} \cdot \tilde{V}(\gamma w;x')\right]$$

$$\tilde{V}^{\pi}(w;x) = E\left[e^{iwR}\, \tilde{V}^{\pi}(\gamma w;x')\right]$$

Discretization frequencies

K = nb points (64, 128, 256)

3 density region (linspace) linear interpolation

Inner region : (0-10]% range (2w) for 50% K

Middle " : (10-50] " " " 30% "

Tail " : (50-100] " " " 20 "

Polar interpolation

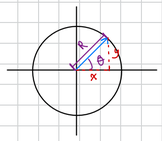
when calculation $\tilde{V}(\gamma w;x')$, γw will probably not be in the discret range of frequencies defined under ~~grid = jnp.linspace(-w, w, 2w/K)~~,

gamma · frequency

thus we will need to interpolate since $\tilde{V}_{\theta^*}(x')$ will output $K \times 2$, based on discrete grid Ω, thus we will need to polar interpolate to find the true $\tilde{V}_{\theta^*}(\gamma w;x')$ within $\tilde{V}_{\theta^*}(w;x')$ which was generated

Polar coordinates (R, θ)

Magnitude, Phase/Angle



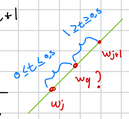
Magnitude $= \sqrt{Re^2 + Im^2}$

Phase $= \arctan\left(\frac{y}{x}\right)$

given $w_q = \gamma w_i$

$w_i < w_q < w_{i+1}$

$$t = \frac{w_q - w_i}{w_{i+1} - w_i}$$



$$\left[\begin{aligned} \mu_q &= t\,\mu_{j+1} + (1-t)\,\mu_j \\ \theta_q &= t\,\theta_{j+1} + (1-t)\,\theta_j \end{aligned}\right]$$

1. calculate $\tilde{V}_{\theta^*}(x')$ for discrete grid $w \in \Omega$
2. Interpolate K points from w to $(\gamma \cdot w)$
3. Calculate loss with $\tilde{V}_{\theta}(x)$ and interpolated $\tilde{V}_{\theta^*}(x')$

Theory	Practice (problem)	Practice (solution)
$\tilde{V}(w;x) \to Re, Im$ ∈ Complex	If we model the NN $\tilde{V}_\theta$ to take as argument both w and x, this means that during complex TD(0) loss, we will need to make K forward pass in $\sum_{w \in \Omega}^{K} \tilde{V}_\theta$ which would be a bottleneck in training	$\tilde{V}_\theta(x) \to$ [Re Im; w_1, w_2, w_3, ⋮, w_K] $K \times 2$ This way, the loss calculation only requires a single forward pass per transition sample

Missing links

- Fast Fourier Transform and its inverse (why does it give so bad results)

TD(0) complex loss

$$L(x, x', \theta, \theta^*, w(\cdot)) = \int_{w \in \Omega} \left| \overbrace{e^{iwR}\, \tilde{V}_{\theta^*}(\gamma w;x')}^{\text{to target} \to \tilde{T}\tilde{V}(x')} - \tilde{V}_\theta(w;x) \right|^2 \cdot w(w)\, dw$$

target network

Weyth function of frequency range

Collapse function

$$\frac{d}{dw}\tilde{V}(w) = \frac{d}{dw} E\left[e^{iwG}\right]$$
$$= E\left[\frac{d}{dw} e^{iwG}\right]$$
$$= E\left[e^{iwG} \cdot iG\right]$$
$$\left.\frac{d}{dw}\tilde{V}(w)\right|_{w=0} = E\left[e^{i \cdot 0 \cdot G} \cdot iG\right]$$
$$= E\left[iG\right]$$
$$\left.\frac{d}{dw}\tilde{V}(w)\right|_{w=0} = i \cdot E[G]$$

$-i$

$$\left[\begin{aligned} V^{\pi}(x) &= E\left[G^{\pi}(x) \mid x = X\right] = -i \frac{d}{dw}\tilde{V}^{\pi}(w=0;x) \\ V^{\pi}(x) &= E\left[G^{\pi}(x) \mid x = X\right] = -\,Im\left(\frac{d}{dw}\tilde{V}^{\pi}(w=0;x)\right) \end{aligned}\right]$$

*2

Final Link

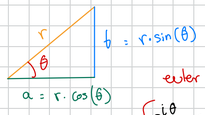
magnitude, phase

$$\tilde{V}(w) = M(w)\, e^{i\theta(w)}$$

CF must respect 2 conditions at $w = 0$

- $M(w=0) = 1$
- $\theta(w=0) = 0$

Polar for complex number



euler formula $\left[e^{i\theta} = \cos(\theta) + i\sin(\theta)\right]$

$$z = a + ib$$
$$= r(\cos(\theta) + i\sin(\theta))$$
$$= \underbrace{r}_{\text{Magnitude}} e^{i\overbrace{\theta}^{\text{Phase}}}$$

Product rule

$$\frac{d\tilde{V}(w)}{dw} = \frac{dM(w)}{dw} \cdot e^{i\theta(w)} + M(w) \cdot \frac{de^{i\theta(w)}}{dw}$$

$$\left.\frac{d\tilde{V}(w)}{dw}\right|_{w=0} = \frac{dM(0)}{dw} \cdot e^{i\theta(0)} + M(0) \cdot e^{i\theta(0)} \cdot i\frac{d\theta(0)}{dw}$$
$$= M'(0) \cdot 1 + 1 \cdot 1 \cdot i\theta'(0)$$
$$= M'(0) + i\theta'(0)$$

$$V^{\pi}(x) = E\left[G^{\pi}(x) \mid x = X\right] = -i \left.\frac{d\tilde{V}(w)}{dw}\right|_{w=0}$$

$$\left.\frac{d\tilde{V}(w)}{dw}\right|_{w=0} = -i\left[M'(0) + i\,\theta'(0)\right]$$
$$= -iM'(0) - i^2\theta'(0)$$
$$= \theta'(0) - iM'(0)$$

*

$$E[G] = Re\left[\theta'(0) - iM'(0)\right]$$
$$= \theta'(0)$$

Numerical Derivative

Finite Difference : $\frac{y_2 - y_1}{x_2 - x_1} = \frac{f(x+h) - f(x)}{h}$

In our case : $-Im\left(\frac{\tilde{V}_\theta(0+h) - V_\theta(0)}{h}\right) = -Im\left(\frac{\tilde{V}(w=h)}{h}\right)$

[1 + 0j] parity of the PDF at $w = 0$

Single point of failure, the Q extraction depends entirely on the precision of a single forward pass

Instead of taking a single points we will use multi-point smoothing using a regression based on a locally Gaussian distribution assumption

Given the CF of a random variable $X \sim N(\mu, \sigma^2)$, then its CF can be written as followed

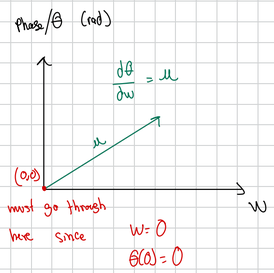
$$C_X(w) = e^{i\mu w - \frac{1}{2}\sigma^2 w^2}$$
$$= e^{-\frac{1}{2}\sigma^2 w^2} \cdot e^{i\mu w}$$
$$= e^{-\frac{1}{2}\sigma^2 w^2} \cdot (\cos(\mu w) + i\sin(\mu w))$$
$$= \underbrace{e^{-\frac{1}{2}\sigma^2 w^2}\cos(\mu w)}_{Re\ (x)} + \underbrace{i\, e^{-\frac{1}{2}\sigma^2 w^2}\sin(\mu w)}_{Im\ (y)}$$

$$\theta = \arctan\left(\frac{y}{x}\right) = \arctan\left(\frac{\cancel{e^{-\frac{1}{2}\sigma^2 w^2}}\sin(\mu w)}{\cancel{e^{-\frac{1}{2}\sigma^2 w^2}}\cos(\mu w)}\right)$$
$$= \cancel{\arctan}\left(\cancel{\tan}(\mu w)\right)$$
$$\theta = \mu w$$

$$\theta(w) = \mu w$$

$f(w) = m \cdot w$ → slope

So for a r.v which is Gaussians the phase/angle is simply $\theta = \mu w$



Ordinary Least Square

$$Q = \mu = \frac{\theta(w)}{w}$$

→ torch.angle (real, imaginary) from the $\tilde{V}_\theta$ network

too much noise on a single point

$$\text{thus } Q = \sum_{w \in \Omega_{collapse}} \frac{\overset{x}{w} \cdot \overset{y}{\theta(w)}}{\underset{x^2}{w^2}} = \mu$$

Single final scalar

if $y = mx$

then Error $= \hat{y} - y$

$$\text{Loss} = \sum(\hat{y} - y)^2$$
$$= \sum_{x \in D}(xm - y)^2$$
$$\frac{\partial L}{\partial m} = \sum 2x_i(x_i m - y_i)$$
$$0 = \quad "$$

$$0 = \sum(x_i^2 m - y_i x_i)$$
$$\sum y_i x_i = m \sum x_i^2$$
$$m = \frac{\sum y_i x_i}{\sum x_i^2}$$